EX. 1. Find an IP over $\mathbb{F}^{m\times n}$ $\quad (A, B) = tr(B^T A)$

2. Let V, W be IPS. Find an IP over $V\times W$, V^*

Furthermore, if N is a subspace of V, V/N

3. Find IP over $\mathbb{R}[[x]]$

2. ① $((v, w), (v', w')) = (v, v') + (w, w')$

② $\dim V = n < +\infty$ $\quad v_1 \cdots v_n$

V^* dual basis $v_1^* \cdots v_n^* \longrightarrow (v_i^*, v_j^*) = (v_i, v_j)$

<u>Application</u> Riesz Representation Thm (家斯表示定理)

Let V be an f.d. IPS / $\mathbb{R}$. Then there is a natural isomorphism between V and its dual V^*

pf. $\sigma: V \longrightarrow V^*$

$v \longmapsto \sigma(v) : u \longrightarrow \sigma(v)(u) = (v, u)$

It's enough to show that σ is inj $(\dim V = \dim V^*)$

$v \in \ker\sigma$ i.e. $\sigma(v) = 0$, so $\forall u \in V$, $(v, u) = 0 \Rightarrow v = 0$

Multiplication of vector spaces

Question $V, W / \mathbb{F} \overset{\text{def}}{\Longrightarrow} T$ s.t. $\dim T = \dim U \times \dim V$

EX $\dim \mathrm{Hom}(V,W) = \dim \mathbb{F}^{\dim W \dim V} = \dim W \times \dim V$

Try to define $V \boxtimes W = \mathrm{Hom}(V,W)$ (not bad, but not natural)

Physics. $V \boxtimes W = \mathrm{span}\{(v_i, w_j)\}$

ENG Quotient (Orthogonal projection)

$V = M \oplus M^{\perp} \xrightarrow{\text{quotient}} M \cong V/M^{\perp}$

Step 1. Construct the free space $F(V \times W)$

EX. Let X be any set. The free space on X is

$F(X) = \mathrm{span}\, X = \{ v \in F(X) \mid v = \sum_{i \in I} a_i x_i,\ a_i \in \mathbb{F}, x_i \in X \}$

step 2. $T = V \boxtimes W = F(V \times W)/N$. what's N?

$$N = \mathrm{span}\left\{ \begin{array}{l} (v,w) + (v',w) - (v+v', w) \\ (v,w) + (v,w') - (v, w+w') \\ \lambda(v,w) - (\lambda v, w),\ \lambda(v,w) - (v, \lambda w) \end{array} \right\}$$

so in $F(V \times W)/N$, $\lambda[(v,w) + N] = (\lambda v, w) + N = (v, \lambda w) + N$

$(v,w) + N = v \otimes w \qquad \lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$

EX. $\mathbb{R}[x,y] = \mathbb{R}[x] \otimes \mathbb{R}[y]$

EX. $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$, $\mathbb{R} \otimes \mathbb{R} = F(\mathbb{R} \times \mathbb{R})/N$, $N =$

Conclusion, $\dim \mathbb{R} \otimes \mathbb{R} = 1$

- Universal Construction (泛构造, 万能构造)

step 1. Bilinear map

$V \times W \xrightarrow{t} T$

$(v, w) \longmapsto t(v,w)$

$(v+v', w) \longmapsto t(v+v', w) = t(v,w) + t(v',w)$

$(v, w+w') \longmapsto t(v, w+w') = t(v,w) + t(v,w')$

$(\lambda v, w) \longmapsto t(\lambda v, w) = \lambda t(v,w)$

$(v, \lambda w) \longmapsto t(v, \lambda w) = \lambda t(v,w)$

$$\begin{tikzcd} V \times W \arrow[r, "t"] \arrow[dr, "\forall b"'] & T \arrow[d, dashed, "\exists! u"] \\ & X \end{tikzcd}$$

$\forall$ Bil t, b $\exists!$ $u \in \mathrm{Hom}(T, X)$ s.t. $b = ut$

Then (T, t) is the tensor product of V and W
$V \otimes W$

<u>Thm</u> Tensor Product is unique up to isomorphism

pf.
$$\begin{tikzcd} V \times W \arrow[r, "t"] \arrow[dr, "t'"'] & T \arrow[d, dashed, shift right, "\exists! u"'] \\ & T' \arrow[u, dashed, shift right, "\exists! u'"'] \end{tikzcd}$$

$I_V = uu' : T' \to T'$

$I_V = u'u : T \to T$